

# **LOMBA KOMPETENSI SISWA PENDIDIKAN MENENGAH TINGKAT WILAYAH JAKARTA TIMUR 2 TAHUN 2026**



## **TEST PROJECT LINUX**

### **IT NETWORK SYSTEMS ADMINISTRATION**

# Introduction

This Linux environment simulates a real-world enterprise network integrating internal services (identity, DNS, certificates), a secured DMZ (web, email), and VPN access. Participants will configure a central firewall to enforce traffic rules, validating proper network segmentation, security, and reliable access for remote clients.

## Login

Username: root/user, Password: Skills39!Jakarta

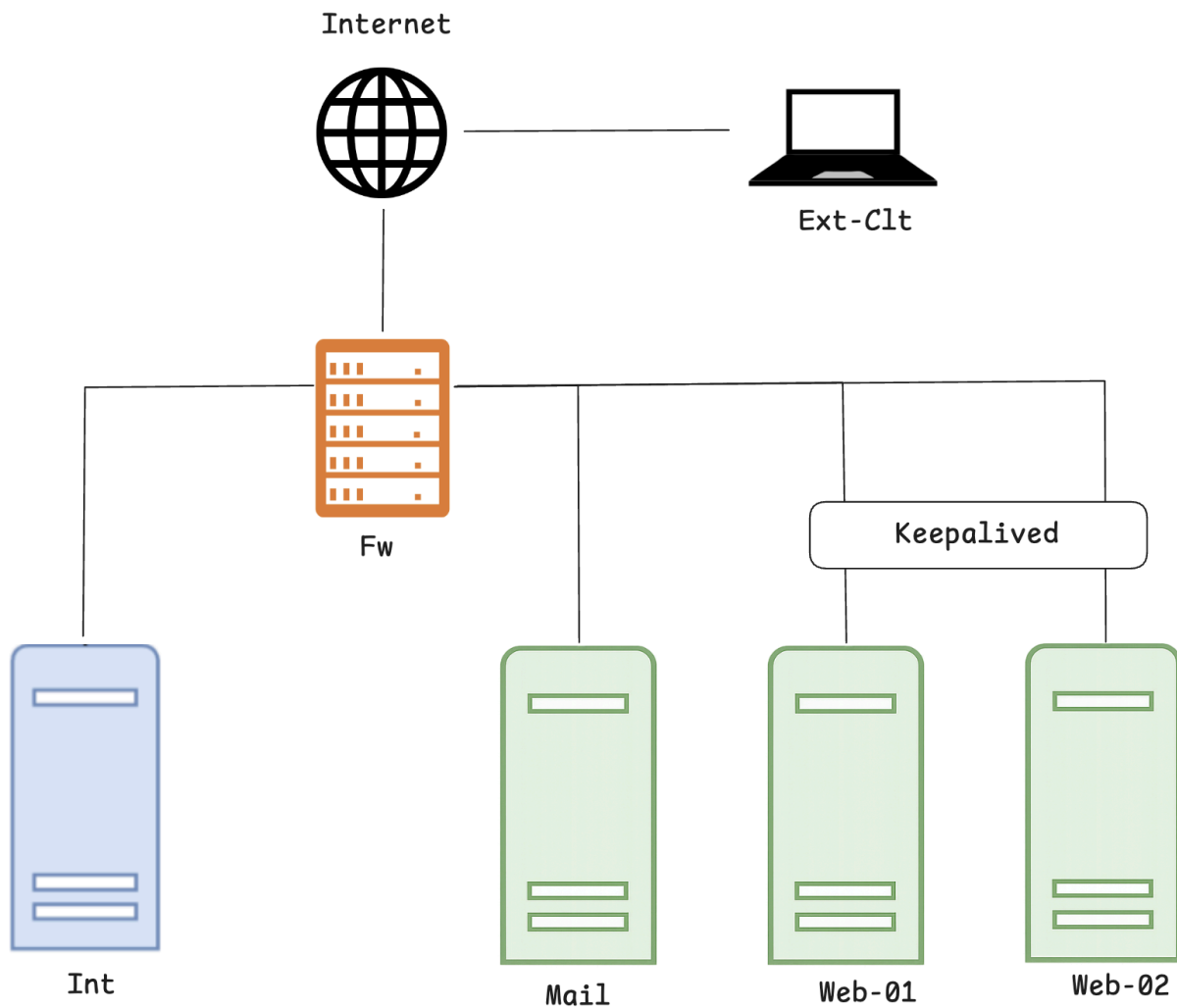
## System Configuration

Timezone: Asia/Jakarta

## General Configuration

| FQDN                  | IPv4                                                                            | Services                                                                    |
|-----------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| fw.lksjakarta.id      | Internet: 200.200.200.100/24<br>Internal: 10.10.10.1/24<br>DMZ: 192.168.26.1/24 | Firewall (nftables), VPN Server (OpenVPN)                                   |
| int.lksjakarta.id     | Internal: 10.10.10.10/24                                                        | DNS Server (Bind9), LDAP Directory (slapd), Certificate Authority (OpenSSL) |
| mail.lksjakarta.id    | DMZ: 192.168.26.10/24                                                           | Mail Server (Postfix(SMTP), Dovecot (IMAP)), WebMail (RoundCube)            |
| web-01.lksjakarta.id  | DMZ: 192.168.26.100/24                                                          | Virtual IP (keepalived), Load Balancer (HAProxy), Web Server (NGINX)        |
| web-02.lksjakarta.id  | DMZ: 192.168.26.200/24                                                          | Virtual IP (keepalived), Load Balancer (HAProxy), Web Server (NGINX)        |
| ext-clt.lksjakarta.id | Internet: 200.200.200.200/24                                                    | Client                                                                      |

## Topology



## Notes

- Ensure that your configuration is saved persistently, any changes must remain intact after a server reboot.
- For the assessment process, the system depends on the SSH service. Therefore, do not modify the SSH configuration under any circumstances!

## Part 1 INT (Internal)

The internal server int.lksjakarta.id is responsible for providing core infrastructure services, including DNS, LDAP authentication, and certificate authority (CA) functions. This setup ensures centralized user management and secure communication across the network.

### int.lksjakarta.id

#### DNS

Set up a DNS server using BIND9 to manage domain names.

1. Install & Configure BIND9
2. Create DNS Zones
  - a. Create a forward zone for: lksjakarta.id
  - b. Create a reverse zone for the int-zone and dmz-zone
3. Add DNS Records
  - a. Add forward and reverse records for all servers listed in the general configuration.
  - b. Add additional records on the table.

| Type  | Name    | Value/Target          |
|-------|---------|-----------------------|
| A     | www     | 200.200.200.100       |
| A     | mail    | 192.168.26.10         |
| A     | vip     | 192.168.26.123        |
| A     | vpn     | 200.200.200.100       |
| CNAME | www.int | vip.lksjakarta.id     |
| MX    | @       | 10 mail.lksjakarta.id |

## LDAP

Install and configure LDAP using slapd to manage user accounts. These users will be used to log in to the mail server.

1. Install LDAP and use slapd
2. Create LDAP users
  - a. Create the following users in LDAP. These users will be used for email login (authentication)

| Full Name                    | Username (uid)               | Password         | Organization Unit (OU) | Email Address                                                          |
|------------------------------|------------------------------|------------------|------------------------|------------------------------------------------------------------------|
| Administrator                | admin                        | Skills39!Jakarta | -                      | -                                                                      |
| Bang Jago                    | jago                         | Skills39!Jakarta | mail, vpn              | jago@lksjakarta.id                                                     |
| kyw1<br>kyw2<br>...<br>kyw10 | kyw1<br>kyw2<br>...<br>kyw10 | Skills39!Jakarta | mail                   | kyw1@lksjakarta.id<br>kyw2@lksjakarta.id<br>...<br>kyw10@lksjakarta.id |
| vpn1<br>vpn2<br>vpn3         | vpn1<br>vpn2<br>vpn3         | Skills39!Jakarta | vpn                    | -                                                                      |

## CA

Use OpenSSL to create your own Certificate Authority (CA) and generate certificates for your server (web and mail).

1. Create Root CA
  - a. Make a Root CA with the common name: "LKSJT2-CA"
  - b. Add fields:
    - i. C = ID
    - ii. ST= Jakarta
    - iii. O = ITNSA
  - c. Install LKSJT2-CA on every server and client
2. Create Certificates for Servers
  - a. Generate and sign the following certificates directly using the Root CA:
    - i. A certificate for web server:
      1. Valid for: www.lksjakarta.id, www.int.lksjakarta.id
      2. Valid for 30 days
    - ii. A certificate for mail server:
      1. Valid for mail.lksjakarta.id
      2. Valid for 60 days
    - iii. A certificate for vpn server has extended Key Usage serverAuth
3. Save All Certificates
  - a. Save the certificates in this folder: /opt/grading/ca
    - i. ca.pem: Root CA certificate
    - ii. web.pem: Web server certificate
    - iii. mail.pem: Mail server certificate
    - iv. vpn.pem: OpenVPN server certificate

## Ansible

We will set up Ansible on the server int and make it control server web-02. All Ansible files (settings, playbooks, and modules) must be saved in the /etc/ansible directory.

1. Configure the Ansible inventory to include web-02, so you can test the connection using the following command: `ansible web-02 -m ping`
2. Create playbook 01-web-int.yml
  - a. This playbook will install common packages like htop, curl, jq and Nginx
  - b. It will also create a virtual host that listens on port 8088
  - c. When you open the website, it will show the message: "This is web internal"
  - d. Make the playbook idempotent
  - e. To run the playbook, use this command:  
`ansible-playbook /etc/ansible/01-web-int.yml`

## Part 2 DMZ (Demilitarized Zones)

The DMZ zone hosts public-facing services such as the mail server and high-availability web servers. These servers are isolated from internal systems but must securely communicate with them for functions like LDAP authentication and certificate verification.

mail.lksjakarta.id

### Mail Server

Deploy a fully functional mail server using Postfix and Dovecot to allow users to send and receive emails under the domain lksjakarta.id.

1. Software Installation
  - a. Postfix - to manage sending and receiving emails SMTP.
  - b. Dovecot - to enable users to access and read emails via IMAP.
2. Mail Server Configuration
  - a. Configure the server to handle mail delivery for the domain lksjakarta.id
  - b. Users should be able to read their emails using IMAP
  - c. Enforce TLS encryption for all connections between email clients and the server
    - i. If the Certificate Authority (CA) task has been completed, use the certificate issued for the mail server.
    - ii. If not, generate a self-signed certificate.
  - d. Ensure clients that trust the LKSJT2-CA can validate the certificate.
3. LDAP Integration
  - a. Integrate with LDAP to authenticate users via int.lksjakarta.id.
  - b. Set up Maildir directories for the following users:
    - i. Jago
    - ii. kyw1 through kyw10
  - c. Ensure that the LDAP user jago can
    - i. Login in using their UID (username).
  - d. Test: use mailutils to confirm that user jago can send an email to themselves.
  - e. Restrict login access:
    - i. Only users under ou=mail are permitted to authenticate and use the mail server.
4. Webmail Access
  - a. Install Roundcube using Apache2 as the web server
  - b. Expose the Roundcube webmail interface on:
    - i. <https://mail.lksjakarta.id>

## web-01.lksjakarta.id & web-02.lksjakarta.id

### HA Web

Set up a high-availability reverse proxy system using HAProxy and Keepalived on two servers: web-01 and web-02.

1. Keepalived Configuration (Virtual IP)
  - a. Create a virtual IP (192.168.26.123/24) that will automatically move to the backup server if the main server goes down.
  - b. web-01: acts as the MASTER with priority 100
  - c. web-02: acts as the BACKUP with priority 90
2. HAProxy Configuration for Load Balancing
  - a. Install and configure HAProxy on both servers (web-01 and web-02) to distribute traffic to two Nginx web servers:
    - i. web-01: 192.168.26.100:8080
    - ii. web-02: 192.168.26.200:8080
  - b. All HTTP requests must be redirected to HTTPS.
  - c. HAProxy should perform TLS termination (HAProxy handles HTTPS and forwards to Nginx using HTTP).
  - d. Add an HTTP header via-proxy: hostname to responses (replace “hostname” with the proxy’s hostname) to help with troubleshooting.
  - e. If you already have a certificate from your CA, use the web certificate (web.pem).
    - i. Make sure a client that only trusts your root CA can validate the entire certificate chain.
    - ii. If not, create a self-signed certificate.
3. Web Server nginx
  - a. On web-01 and web-02 install nginx and listen port 8080 with index.html “Hello from \${hostname}”

## nftables

Create firewall and NAT rules using nftables to control which traffic is allowed or blocked in your network.

1. Allow Internet Access
  - a. Devices in the INT (internal) and DMZ networks can access the Internet area.
2. NAT Masquerade
  - a. When sending traffic from your network to the internet (through the WAN), the firewall must replace the source IP with its own WAN IP using masquerade NAT.
3. Port Forwarding
  - a. Forward incoming traffic from the internet to servers in the DMZ:

| Port | Protocol | Forward To     |
|------|----------|----------------|
| 80   | TCP      | 192.168.26.123 |
| 443  | TCP      | 192.168.26.123 |
| 53   | TCP/UDP  | 10.10.10.10    |

4. VPN Access
  - a. VPN clients should be able to reach both internal and DMZ networks.
5. Mail Server to LDAP
  - a. The mail server must be allowed to query the LDAP service running on int.
6. Block Everything Else
  - a. Any other traffic that's not mentioned above should be blocked (default deny).

## OpenVPN

Ensures secure access to internal web applications, email systems, and internal DNS services.

1. Install OpenVPN along with the LDAP authentication plugin
  - a. Set up a Tun interface for Layer 3 tunneling:
    - i. Interface type: tun
    - ii. Uses UDP protocol on port 1194
2. Assigns VPN clients to the 10.1.26.0/24 subnet
3. Use a topology option that utilizes IP space efficiently
4. Utilize a certificate issued by LKSJT2-CA
5. Enable LDAP-based authentication via int.lksjakarta.id
6. Authentication base DN: ou=vpn,dc=lksjakarta,dc=id
7. Configure OpenVPN log to be located in /var/log/openvpn/openvpn.log and make sure the log will be appended.

## Part 4 Client

The client machine **ext-clt.lksjakarta.id** represents a remote workstation that connects securely to internal and DMZ services through VPN. It is used to validate the overall infrastructure by testing access to web, DNS, and email services.

### General Configuration

1. Create a local user named jago (Bang Jago)
  - a. Set the password to Skills39!Jakarta
  - b. Grant jago passwordless sudo privileges
2. Set up the OpenVPN client
  - a. Save the .ovpn configuration file to /etc/openvpn/client/lksjakarta.conf
  - b. Create a file /etc/openvpn/auth.txt to securely store OpenVPN credentials. use user jago on ldap with DN: ou=vpn,dc=lksjakarta,dc=id
  - c. Create a systemd service named openvpn-client.service to enable automatic startup on boot and ensure persistent VPN connectivity
3. Ensure access to internal web services via vpn
  - a. Able to access <https://www.lksjakarta.id> and <https://www.int.lksjakarta.id> without certificate errors when connected to the VPN
4. Verify webmail access through VPN
  - a. Access Roundcube webmail at <https://mail.lksjakarta.id> after VPN connection
  - b. Successfully log in using the account [jago@lksjakarta.id](mailto:jago@lksjakarta.id)
  - c. Able to send an email to the same account (self-mail test) without issue